Qubit Realisation using the Diamond NV-Center

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Abstract

Nitrogen-Vacancy (NV) centres in diamond have emerged as suitable hardware platforms for qubit implementation, for they combine atom-like properties in a solid-state host. The spin of the negatively-charged NV electron and the surrounding Carbon atoms can be manipulated by arbitrary magnetic fields to create coupling and entanglement, which are crucial processes for quantum computing operations. Not only can the negatively charged defect operate at room-temperature, but the readout of the state of the system can be performed optically with high fidelity, which are significant advantages compared to rival platforms and further increase interest in this technology.

1 Introduction

Quantum computing exploits quantum mechanical phenomena such as superposition and entanglement to perform computations fundamentally different from classical computers, promising exponential speedups for specific problems including integer factorisation, database searching, and quantum simulation. However, practical realisation requires physical qubits that can be reliably initialised, manipulated, measured, and scaled whilst maintaining quantum coherence. Multiple platforms have been proposed—superconducting circuits, trapped ions, photonic systems, and solid-state spins—each with distinct advantages and trade-offs. Superconducting qubits offer fast gates but suffer from short coherence times and require cryogenic cooling. Trapped ions excel in coherence and fidelity but face scalability challenges. In this context, the nitrogen-vacancy (NV) centre in diamond has emerged as a compelling alternative, offering room-temperature operation, millisecond coherence times, and high-fidelity optical readout. [1]

The NV centre is a negatively-charged defect in diamond consisting of a nitrogen atom substituting a carbon atom, adjacent to a lattice vacancy. Its remarkable properties include: (i) room-temperature operation eliminating cryogenic requirements, (ii) long spin coherence times ($T_2 \sim$ milliseconds) far exceeding superconducting qubits, (iii) high-fidelity optical initialisation and readout via spin-dependent photoluminescence, and (iv) ability to couple the electron spin to surrounding nuclear spins, enabling multi-qubit registers within a single defect. Recent demonstrations have achieved 10-qubit registers with 7-qubit entanglement and gate fidelities of 99.1%, validating the platform for practical quantum information processing. [7]

This review comprehensively examines the NV centre as a qubit platform: we discuss its photophysical properties and electronic structure (Section 2), detail single-qubit control and coherence measurements (Section 3), address multi-qubit implementations including two-qubit gates and entanglement generation (Section 4), and assess the platform against the DiVincenzo criteria for universal quantum computation (Section 5). Throughout, we emphasise both the remarkable capabilities and remaining challenges, including strategies for improving coherence times and gate fidelities towards fault-tolerant quantum computing.

2 Photophysics of the NV Center

According to L. Rodin et al. [2], the NV defect consists of a combination between a Nitrogen atom (N) impurity, which occupies the position of a Carbon atom (C), coupled to a vacancy (V) in one of the neighbouring sites of the diamond crystal lattice. With C_{3v} symmetry, the defect behaves as an artificial

atom in the diamond matrix, where the nitrogen-vacancy pair is oriented along the [111] crystalline direction. This axis, which joins the nitrogen and the vacancy, serves as the intrinsic quantisation axis, defining the spin projection direction, m_s.

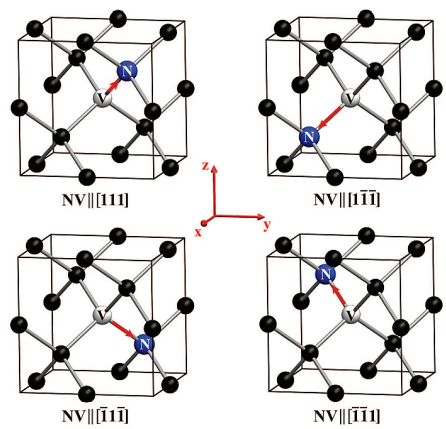


Figure 1: Four possible orientations of the NV defect in diamond. [3]

The NV centre has two forms: a neutral state, NV^0, and a negatively charged state, NV^-. The latter state has more relevance, as the it possesses two unpaired electrons that provide a spin triplet ground level which can be initialised, coherently manipulated and readout by optical means. [2]

2.1 Energy Level Scheme

For simplicity, the NV^- will be refered simply as NV.

2.1.1 Ground and Excited States

As mentioned above, the ground level of the NV centre is a spin triplet state, $^3A_2$. Spin-spin interaction splits its energy into a singlet state, corresponding to spin projection $m_s = 0$, and a doublet state, corresponding to spin projection $m_s = \pm 1$, separated by $D = 2.88$ GHz [4] when no magnetic field is present [2].

The defect can be excited through spin-conserving transitions to the excited state ^3E. Although ^3E is a spin triplet, its orbital doublet is averaged at room temperature. At lower temperatures, spin-orbit coupling becomes significant, leading to orbital anisotropy. Deformities in the crystal lattice break the degeneracy between the two orbitals, causing them to have different energy values. At room temperature, however, thermal excitation enables transitions between the two orbitals to occur rapidly enough that they are effectively averaged. This means that the doublet orbitals become indistinguishable, and the electron's energy is determined solely by its spin state. This leads to a zero-field splitting $D_{es} = 1.42$ GHz along the same quantisation axis and gyromagnetic ratio as in the ground-level.

In addition to $^3A_2$ and ^3E, there are two more singlet states: ^1E and ^1A, which play a significant role in the spin dynamics.

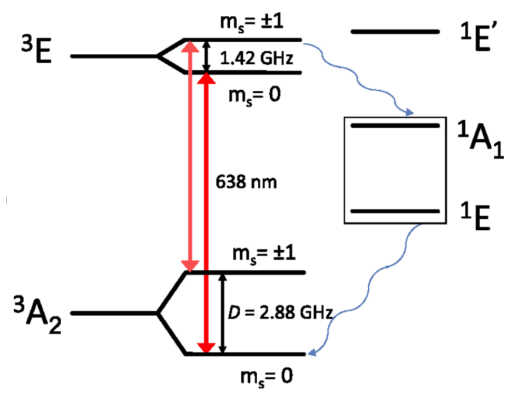


Figure 2: Schematic diagram about energy levels of the NV center at room temperature. [4]

2.2 Electronic Transitions and Photoluminescence

The NV^- center exhibits broadband photoluminescence (PL) with an emission zero-phonon line at 1.945 eV, corresponding to a wavelength of $\lambda_{\text{ZPL}} = 637$ nm, which is one of the optical properties that allow for its readout using completely optical means. There are two main transition mechanism between the excited and ground states: radiative spin-conservative optical transition ($\Delta m_s = 0$), and non-radiative intersystem-crossing (ISC), which is extremely spin-selective.

When the system is excited from the ground level $^3A_2$ to the ^3E level, which is a spin-conserving transition, using non-resonant green illumination on the phonon sidebands ($\lambda = 532$ nm) the relaxation process can occur through a radiative transition to $^3A_2$, producing a broadband red PL ($\lambda = 637$ nm) [2].

Because ISC is a non-radiative process, its detection is done by use of Electron Spin Resonance (ESR). The shelving rate from the $m_s = 0$ sublevel is much smaller than those from $m_s = \pm 1$, making ISCs to the ^1E singlet strongly spin selective. The NV defect preferentially decays from the lowest $^1A_1$ singlet state towards the ground state $m_s = 0$ sublevel. Through optical pumping, it can be seen that these spin-selective processes provide a high degree of electron spin polarisation into $m_s = 0$.

Since only one of the relaxation schemes is radiative, the NV centre's PL intensity is significantly higher when the $m_s = 0$ state is populated. When a resonante microwave (MW) field is applied to single defect, initially in state $m_s = 0$, it is driven to the $m_s = \pm 1$ spin state and a drop in the PL signal is observed.

2.3 Magnetic Field Sensitivity

When magnetic field is applied on the vicinity of the NV centre, the degeneracy of the $m_s = \pm 1$ is lifted by the Zeeman effect, leading the appearance of two resonance lines in the ESR spectrum. This means the magnetic field can be measured by evaluating the detection of Zeeman shifts of its electron spin sublevels.

2.3.1 Ground-state Spin Hamiltonian

The relationship between ESR frequencies and the magnetic field is expressed in the ground state Hamiltonian. Ignoring the hyperfine interaction with nearby nuclear spins the the diamond lattice, we have:

$$\mathcal{H} = hDS_z^2 + hE(S_x^2 - S_y^2) + g\mu_B \boldsymbol{B} \cdot \boldsymbol{S} \tag{1}$$

where z is the NV defect quantisation axis, h is the Planck constant, D and E are the zero-field splitting parameters, S_i the Pauli matrices, $g \simeq 2$ is the Landé factor and μ_B is the Bohr magneton. Parameter $D \approx 2.88$ GHz results from spin-spin interaction between the defect's two unpaired electrons and is higly sensitive to temperature fluctuations. Parameter E is the off-axis zero-field splitting parameter and results from local strain in the diamond matrix, which lowers the C_{3v} symmetry. This value can go from $E \approx 100$ kHz to a few MHz (typically, $E \sim 5$ MHz, but the condition $E \ll D$ is always fulfilled.

Taking into account the magnetic field projection along the NV defect, the Hamiltonian from Eq. 1 can be expanded. With $B_{NV} = B_z$, we then get:

$$\mathcal{H}_{\parallel} = hDS_z^2 + g\mu_B B_{NV} S_z$$
$$\mathcal{H}_{\perp} = g\mu_B (B_x S_x + B_y S_y)$$

Which transforms the Total Hamiltonian into:

$$\mathcal{H} = \mathcal{H}_{\parallel} + \mathcal{H}_{\perp} + hE(S_x^2 - S_y^2) \tag{2}$$

The two ESR frequencies, $\upsilon_{\pm}$, can be calculated my computing the eigenenergies of the Hamiltonian as a function of the magnetic field amplitude, $\|B\|$, for various angles θ between the magnetic field and the NV centre axis.

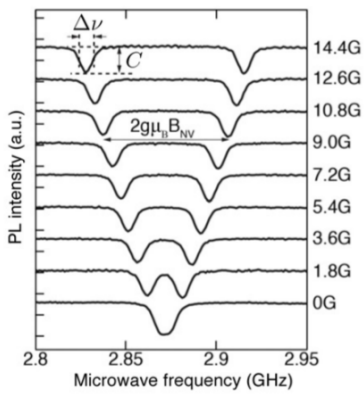


Figure 3: Optically detected ESR spectra for different magnetic field amplitudes applied to a single NV defect. Obtained by monitoring the NV defect PL intensity while performing a microwave field sweep. [2]

In Fig. 3, the ESR transitions are shifted due to the Zeeman effect, providing a quantitative measurement of the magnetic field projection along the NV quantisation axis. The distance between each dip is given by the Zeeman Effect energy shift:

$$\Delta E = 2g\mu_B B_{NV} \tag{3}$$

2.4 Hyperfine Structure

The interaction between the electrons and the nuclei of the system, specially with the Nitrogen nucleus, have to be considered. For a nuclear spin I of the Nitrogen nucleus, each sublevel of the of the ground state, ($m_s = \{0, \pm 1\}$), is split into $2I + 1$ levels. The splittings between these levels are generally considered constant as they have a very small dependence with the magnetic field.

New resonance hyperfine lines are originated for each $\Delta m_s = \pm 1$ transition, totalling up to to $2I + 1$ lines that can be seen in a magnetic resonance experiment. The splitting between these hyperfine lines, A_{hf}, depends only on the Nitrogen isotope of the centre: $A_{hf} = 2.16$ MHz for ^{14}N, and $A_{hf} = 3.05$ MHz for ^{15}N. It is important to notice, however, that ^{14}N represents 99.64% of naturally occurring Nitrogen, meaning most NV centres are composed of this isotope [5].

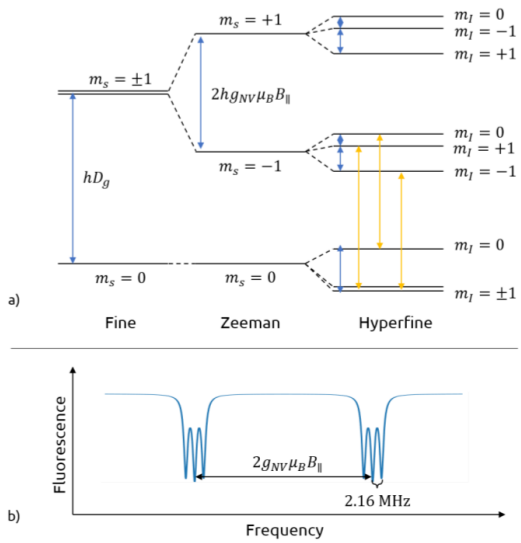


Figure 4: (a) NV ground state configuration for Zeeman Effect and Hyperfine interaction for the ^{14}N isotope. (b) Magnetic Resonance for a magnetic field applied along the NV axis. [5]

3 Qubit Implementation

To study concepts such as the relaxation times T_1 and T_2 and Rabi oscillations, we work with ensembles of NV centres. It is necessary to ensure that a maximum of NV centres are put in the same state simultaneously. The overall signal is proportional to the population difference between the levels. At thermal equilibrium, the occupation ratios, n_i of the three spin states are governed by Boltzmann statistics.

$$\frac{n_1}{n_0} \approx \frac{n_{-1}}{n_0} \approx \exp\left(\frac{-hD}{k_B T}\right) \tag{4}$$

Where k_B is the Boltzmann constant and T is the absolute temperature. For $T = 300$ K, $n_1/n_0 \approx 0.99954$, meaning each state is approximately equally populated by one third of the NV centres and no signal can be extracted from the signal (in section. 2.2 it was explained that PL intensity is significantly higher when the $m_s = 0$ state is populated). It is then necessary to put the system out of equilibrium and into state $m_s = 0$ using optical pumping.

3.1 Spin Manipulation

Since the unpaired electrons' spin is sensitive to external magnetic fields, The spin state can be read by measuring the PL signal of the defect. The NV centre can be polarised by applying a Radio Frequency (RF) excitation corresponding with the difference between the $m_s = 0$ and $m_s = \pm 1$ states. A population transfer between these two spin states causes a drop in the PL signal wich is then optically detected.

3.1.1 Optically Detected Magnetic Resonance (ODMR)

The $m_s = \pm 1$ degeneracy is lifted by the Zeeman Effect in the presence of a magnetic field with a strong component along the NV centre's quantisation axis, $B_\parallel$. As seen in section 2.3.1, the separation between the $m_s = \pm 1$ sublevels is dependant on the magnetic field, being equal to $hg_{NV}\mu_B B_\parallel$, making the NV centre an atomic scale magnetometer. The magnetic resonance spectrum of the NV centre shows 8 lines, which correspond to the four possible transitions (between $m_s = 0$ and $m_s = +1$ and between $m_s = 0$ and $m_s = -1$ states) in the four [111] crystal orientations.

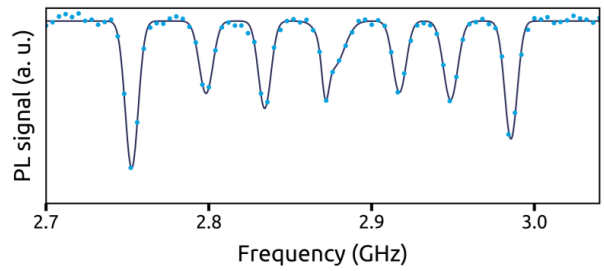


Figure 5: Magnetic Resonance spectrum under the application of a bias field in the order of a few mT.

3.1.2 Longitudinal Relaxation Time, T_1

The longitudinal relaxation time corresponds to the time required for the system to return into thermodynamic equilibrium after being polarised into the $m_s = 0$ spin state. The T_1 time is limited by the interactions between the NV centre and the phonons in the crystal lattice.

The standard sequence for T_1 measurement consists of a first initialisation laser pulse, polarising the system into $m_s = 0$, followed by a free evolution time of variable duration τ. Lastly, a readout laser pulse extracts the spin state of the NV centre by use of its PL signal. It is important to minimise light in the presence of the diamond, so interference with the system is diminished.

The duration τ should be long enough for the system to return to its thermodynamic equilibrium. For short values of τ, such is not possible, which causes the system to remain polarised in the state $m_s = 0$. This results in a high PL signal, which decreases as τ increases and the system gradually relaxes towards equilibrium.

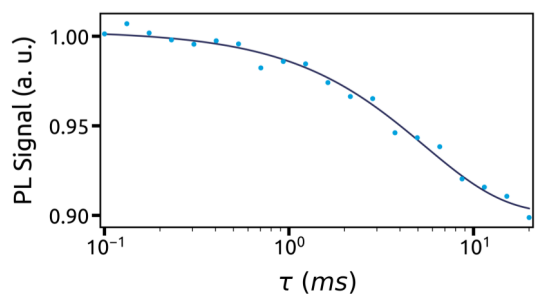


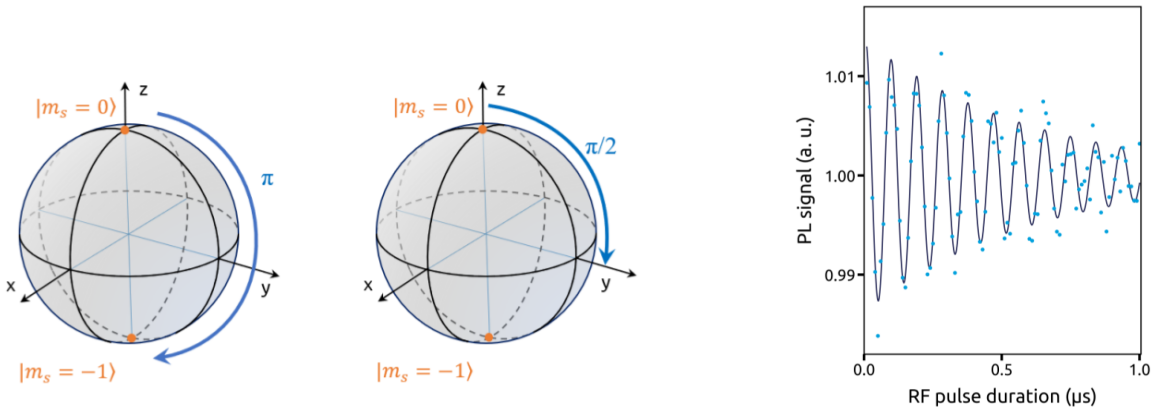
Figure 6: Evolution of the PL signal as a function of the free evolution time, τ. Exponential decay fitting was used to obtain T_1. For an NV centres ensemble in a bulk diamond, a typical value is $T_1 = 5.4 \pm 0.5ms$.

3.1.3 Rabi Oscillations

The coherent control of the NV centre's spin state can be achieved by performing Rabi oscillations between the states $m_s = 0$ and $m_s = +1$. For this last state, a bias magnetic field is applied along the quantisation axis to lift the degeneracy between the $m_s = \pm 1$ states. A possible sequence consists on a first initialisation pulse, polarising the NV centre into the $m_s = 0$ spin state, followed by a RF pulse of variable duration, τ, that is resonant with the transition between states $m_s = 0$ and $m_s = +1$: this is the manipulation step. After this step, the system is in a superposition of states: $\alpha \left| m_s = 0 \right\rangle + \beta \left| m_s = \pm 1 \right\rangle$. Lastly, the spin state of the system is extracted by a readout laser pulse.

3.1.4 Dephasing Time, T_2^*

Environmental noise fluctuations cause loss of phase coherence of an ensemble in a quantum superposition $\alpha \left| m_s = 0 \right\rangle + \beta \left| m_s = \pm 1 \right\rangle$, where the relative phase between states decays. During this process, known as

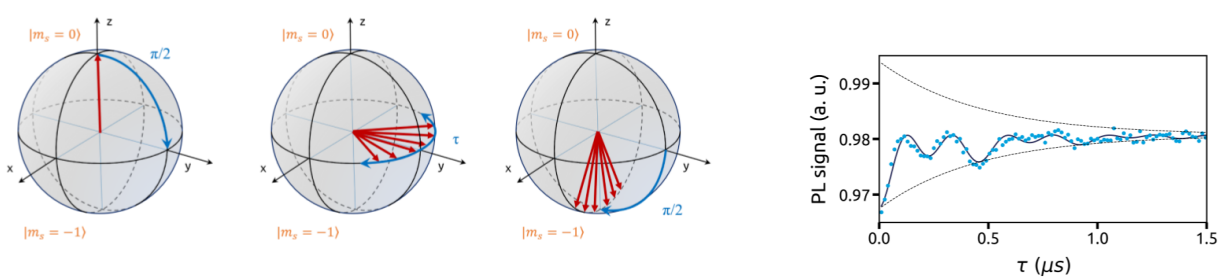


(a) (b)

Figure 7: (a) Bloch Sphere representation of each step of the sequence. (b) Typical Rabi oscillations on an NV centre. The solid line is a sinusoidal function fit with an exponential decay envelope.

dephasing, the system loses quantum behaviour.

The dephasing time, T_2^* can be measured by the Ramsey fringe method: after pumping, a first RF pulse of duration $T_{\pi/2}$ sets the system into a superposition state, then the system freely evolves for a time τ through free precession. Each NV centre accumulates a phase, which not only depend on its environment, they strongly depend on the magnetic field fluctuations of the spin bath. A second RF pulse of duration $T_{\pi/2}$ is applied, followed by the readout laser pulse.



(a) (b)

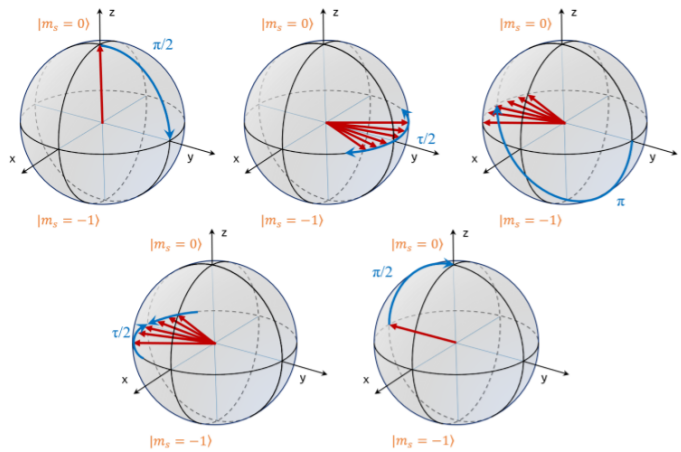
Figure 8: (a) Bloch Sphere representation of each step of the sequence. (b) Measurement of Ramsey fringe on a set on NV centres.

The decay of the oscillations in fig. 8.b) corresponds to the dephasing time, with typical value $T_2^* = 456 \pm 45$ ns. In NV centres, the dephasing is governed by the interactions with the spin bath, which are composed by nuclear spins (such as ^{13}C atoms with nuclear spin $I = 1$) and electronic spins, such as nitrogen impurities or on the diamond's surface. When the system is sufficiently isolated from environmental magnetic noise, the characteristic time, T_2 can be referred as the coherence time, instead of T_2^*

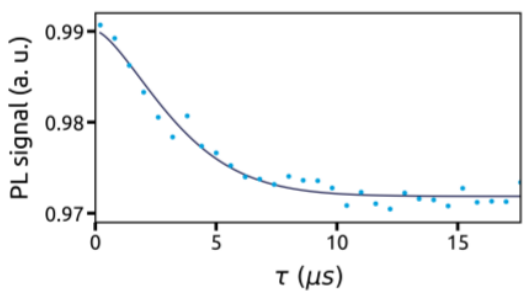
3.1.5 Coherence Time, T_2^{echo}

Coherence time with dynamic decoupling refers to the time duration a quantum system can maintain its superposition state, achieved by actively cancelling out environmental noise through specific, rapid control pulses. The Hahn echo sequence is the simplest dynamic decoupling sequence to implement. This sequence is similar to that of the T_2^* time measurement, except now a RF pulse of duration T_π is applied at the middle of the free precession time τ. The phase shift accumulated during the time $\tau/2$ is compensated by

the phase accumulated by the second free precession time $\tau/2$. This method then rejects slow fluctuations of the environment, protecting the NV centre from noise.



(a)



(b)

Figure 9: (a) Bloch Sphere representation of each step of the sequence. (b) Measurement of coherence time T_2^{echo}. When fitting the data by an exponential decay to the power of 2, one can obtain $T_2^{echo} = 3.8 \pm 0.2\ \mu s$.

4 Multi-Qubit Implementation

This section is written according to C. E. Bradley et al. [7]. Even though spins associated to single defects in provide promising qubits, their control has been limited to only a few qubits. As of now, the maximum number of qubits controlled consists of the electron spin of a NV center and 8 nuclear spins in diamond.

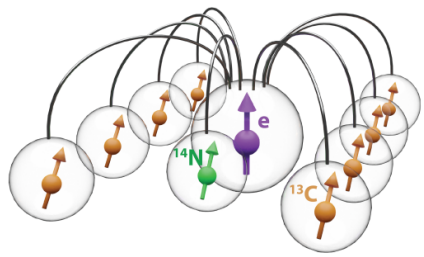


Figure 10: 10-qubit register with the electron spin of a single NV center acting as a central qubit, connected by two-qubit fates to the intrinsic ^{14}N nuclear spin and further 8 ^{13}C nuclear spins surrounding the NV centre. [7]

4.1 Electron and Nuclear Spins

Different types of spins fulfil different types in a quantum computer. Electron spins enable fast control and can be read with high fidelity, from methods such as ESR and MR mentioned above. Further properties can be obtained from coupling electrons to other electrons or to photons. The first, electron-electron coupling, enables on-chip connectivity between defects, while electron-photon coupling allows for the realisation of long-range entanglement links. Nuclear spins can be used to store and process quantum states, serving as additional qubits with long coherence times.

4.2 Multi-qubit Gates

The required gates to realise a multi-qubit register need to selectively control the qubits and, simultaneously, decouple unwanted interactions so coherence can be protected. In order to achieve that, a protocol based on selective phase-controlled driving of nuclear spins interleaved with decoupling sequences on an electron spin can be used. In these hybrid systems, multiple spins are coupled, making it difficult to maintain coherence on the electron spin qubit and to avoid unwanted crosstalk. The previously mentioned unwanted interactions arise from the electron spin continuously coupling to all Carbon spins through the hyperfine interaction, and the dynamics of electron spin and nuclear spins occurring on different timescales. This leads to the necessity of creating decoherence-protected gates that perform decoupling sequences on the electron spin, combined with nuclear spin control.

4.2.1 Two-Qubit Gates: DDRF

Considering an NV centre in diamond and its surrounding nuclear spins. The two-qubit gates use electron-nuclear interaction to control the Nitrogen spin and several Carbon atoms' spin. A possible two-qubit gate implementation scheme, DDRF gate, is based upon phase-controlled radio frequency (RF) driving of nuclear spins, interleaved with dynamical decoupling (DD) of the electron spin. The hyperfine interaction, which couples each nuclear spin to the electron spin, depends on the relative position of the spin to the NV centre. Because of this, different nuclear spins can be distinguished by their precession frequencies. In the interaction picture with respect to the electron energy splitting, and neglecting non-secular terms, the Hamiltonian describing the electron and a single C nuclear spin is given by: [8, 9, 10]

$$\mathcal{H} = \omega_L I_z + A_{\parallel} S_z I_z + A_{\perp} S_z I_x \tag{5}$$

Where $\omega_L = \gamma B_z$is the nuclear Larmor frequency (set by the external magnetic field B_z along the NV axis), γ is the ^{13}C gyromagnetic ratio, S_i and I_i are the spin-1 and spin-1/2 operators of the electron and nuclear spins, respectively, and $A_{\parallel}$ and $A_{\perp}$ are the parallel and perpendicular hyperfine components. The phase, ϕ, Rabi frequency, Ω, and frequency, ω parameters of an RF pulse can be set to control a target nuclear spin. To target a specific nuclear spin, we set $\omega = \omega_1$, where ω_1 is the spin precession frequency when the electron is in the $m_s = -1$ spin projection, with $\omega_1 = \sqrt{(\omega_L - A_{\parallel})^2 + A_{\perp}^2}$. Assuming $(\omega_L - \omega_1) \gg \Omega$, the driving of the nuclear spin is negligible while the electron is in the $m_s = 0$ spin projection. Parameter

$A_\perp$ can be set as zero for simplicity. Considering only the $m_s = \{0,\ -1\}$ subspace, with the addition of RF driving and in a rotating frame at the RF frequency, the Hamiltonian of Eq.5 becomes: [7, 11]

$$\mathcal{H} = |0\rangle\langle 0| \otimes (\omega_L - \omega_1) I_z + |1\rangle\langle 1| \otimes \Omega(\cos(\phi) I_x + \sin(\phi) I_y) \tag{6}$$

where $|0\rangle$ and $|1\rangle$ represent the electron $m_s = 0$ and $m_s = -1$ spin projections, respectively. In this picture, the state of the electron defines the direction of the nuclear spin. When the electron is in state $|0\rangle$, the nuclear spin undergoes procession around the z-axis at frequency $(\omega_L - \omega_1) = A_\parallel$. While the electron is in state $|1\rangle$, conversely, the nuclear spin is driven around a rotation axis in the x-y-plane, with an angle defined by the phase of the RF field, ϕ.

To simultaneously decouple the electron spin from the environment, the RF pulses are interleaved in a sequence of the form:

$$(\tau - T_\pi - 2\tau - T_\pi - \tau)^{N/2}$$

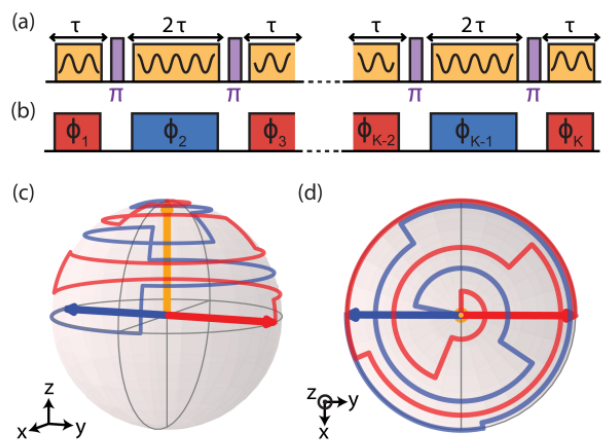


Figure 11: (a) Pulse sequence employed to realise a DDRF gate. Dynamical Decoupling pulses on the electron spin (purple) interleaved with RF pulses (yellow) which selectively drive a single nuclear spin. (b) RF pulses resonant with the nuclear spin (with the electron in state $|0\rangle$, blue, and in $|1\rangle$, red) at the start if the two-qubit gate. The phase of each RF pulse is adapted to create the desired nuclear spin trajectory account for periods of free precession. (c) Example of nuclear spin evolution on the Bloch Sphere with $N = 8$ electron decoupling pulses. Starting from initial state $|\uparrow\rangle$ (yellow), the blue (red) path shows the nuclear spin evolution fir the case when the electron starts in $|0\rangle$ ($|1\rangle$). He final state vectors are anti-parallel along the equator, meaning the gate is a maximally entangling two-qubit gate. (d) Top-down view of the process described in c). [7]

where 2τ is the inter-pulse delay and N is the total number of electron decoupling pulses.

4.2.2 Definition of the RF phase

The evolution of the nuclear spin during this sequence is considered separately for the two initial electron eigenstates: $|0\rangle$ and $|1\rangle$. Each successive pulse is labelled by an integer $k = 1, .., K$, where $K = N + 1$ is the total number of RF pulses. The initial electron state influences the value of k. This means that the desired nuclear spin evolution can be created by setting the phases of the Rf pulses, ϕ_k.

- If the initial electron spin is in state $|0\rangle$, only the even k RF pulses will be resonant.
- If, however, it is in state $|1\rangle$, only the odd k pulses are resonant.

To have the RF sequential rotations building constructively, the phases of reach RF pulse are set to account for the periods of nuclear spin precession between them, which build up in integer multiples of $\phi_\tau = (\omega_L - \omega_1)\tau$. The RF pulse phases are summarised by $\phi_k = \varphi + \phi'_k$, where:

$$\phi_k = \begin{cases} (k-1)\phi_+ + \pi & k \text{ odd} \\ (k-1)\phi_- & k \text{ even} \end{cases} \tag{7}$$

This means we have:

- When the electron starts in $|0\rangle$ and k is even, the required phase sequence is $\phi_\tau, 3\phi_\tau, 5\phi_\tau ...$
- When it starts in $|1\rangle$ and k is odd, the required phase sequence is $0, 2\phi_\tau, 4\phi_\tau ...$

4.2.3 Conditional and Unconditional Rotations

An unconditional rotation, corresponding to a single-qubit gate, and a conditional rotation, corresponding to a two-qubit gate, can be constructed. In Eq. 7, the optional π phase shift for the odd sequence converts the unconditional rotation into a conditional rotation, while adding a further phase φ to all pulses allows to set the rotation axis of the gate. Having in mind the choice of phases above, the total evolution of the two-qubit system is given by $V = V_z \cdot V_{CROT}$. Here, V_z is an unconditional rotation of the nuclear spin around the z-axis and V_{CROT} is a conditional rotation of the nuclear spin, which depends on the electron state.

$$V_{CROT} = |0\rangle\langle 0| \otimes R_\varphi(N\Omega\tau) + |1\rangle\langle 1| \otimes R_\varphi(-N\Omega\tau) \tag{8}$$

With $R_\varphi(\theta) = e^{-i\theta(\cos(\varphi)I_x + \sin(\varphi)I_y)/\hbar}$. After setting up parameters (N, Ω and τ, for the rotation angle, and φ for the rotation axis), V_{CROT} describes a controlled rotation of the nuclear spin with tuneable rotation angle and rotation axis. Setting $(N\Omega\tau) = \pi/2$ achieves a maximally entangled two-qubit operation equivalent to a $CNOT$ gate up to local operations. Example dynamics for a nuclear spin evolving under such a sequence is shown in Fig. 11 (c) and (d).

This design for a DDRF gate is able to provide average gate fidelity $\mathcal{F}_{gate} = 0.991$. This high result can be explained due to various properties of the DDRF protocol: it allows for nuclear spins with small or negligible $A_\perp$ can be controlled, increasing the number of accessible nuclear spin qubits. In addition to that, because the targeted dynamics are achieved by setting the RF phases and amplitudes, the inter-pulse delay τ of the DD sequence can be freely optimised to protect electron decoherence. It can also be used for selective control of nuclear spins coupled to spin-1/2 defects. via a contact hyperfine coupling. The use of an RF field to control the qubits opens promising possibilities like parallelising gates by frequency multiplexing and using shaped and composite pulses to mitigate dephasing and crosstalk, thus improving coherence. [7]

4.2.4 Producing GHZ Entangled States

An important requirement for a quantum processor to be able to perform operations on its multiple constituents within a single algorithm. Generating GHZ type states serves as a way to verify this capability. A N-qubit GHZ state is defined as:

$$|GHZ_N\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |+\rangle^{\otimes(N-1)} + |1\rangle \otimes |-\rangle^{\otimes(N-1)}\right) \tag{9}$$

Characterising the complete quantum state of large multi-qubit systems is computationally prohibitive due to the exponential growth of the Hilbert space. However, the quantum entanglement of all N qubits can be efficiently verified using entanglement witnesses with reduced measurement bases. For GHz states with system size 2^N, the density matrix can be reconstructed via linear inversion from 2^N measurement operators, enabling fidelity assessment with the target state. The entanglement witness value $\mathcal{W}_{\text{GHZ}} = \mathbb{1}/2 - |GHZ_N\rangle\langle GHZ_N|$ indicates genuine multipartite entanglement. To achieve the required measurement bases, we employ multi-qubit parity measurements of Pauli operators on the register, with the readout sequence adapted for the electron spin state by rotating the nitrogen nucleus such that the desired basis aligns with the Z-axis. This approach protects the measured population from dephasing induced by optical readout of the electron spin.

As the number of qubits increases, maintaining nuclear spin coherence becomes challenging because the total sequence time becomes comparable to the natural dephasing time (T_2) of the nuclear spins. To preserve

nuclear spin coherence during extended sequences, we insert Hahn RF pulses (section 3.1.5) at regular intervals to refocus the nuclear spin state at the point of the next operation.

Genuine N-qubit entanglement of up to 7 qubits has been achieved high statistical significance. Fidelity measures with target GHZ states for 2 to 8 qubits, with $N = 5$ and $N = 7$ qubits have yielded fidelities 0.804 and 0.589, respectively. [7]

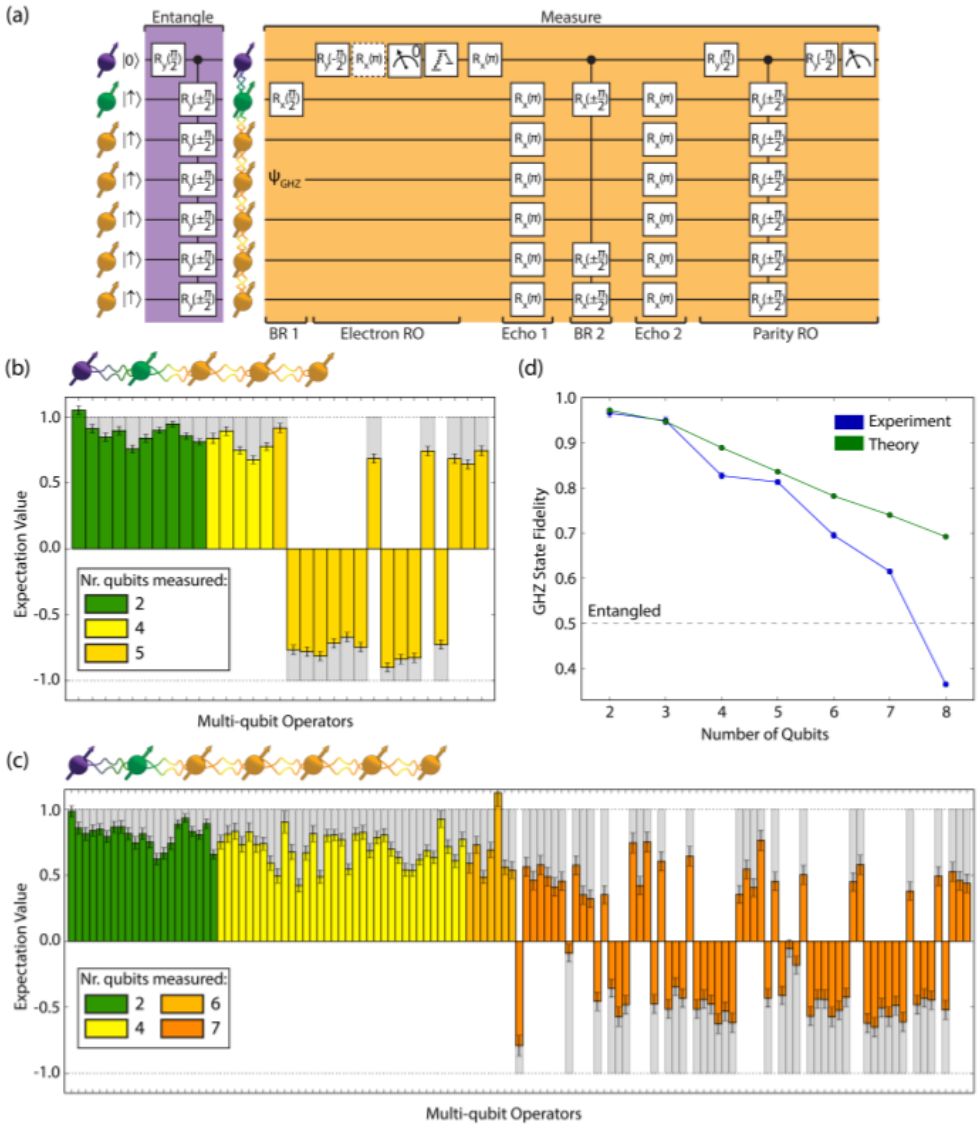


Figure 12: (a) Experimental sequence for the preparation of a 7-qubit GHZ state (purple) and measurement of operator $XYYYYZZ$ (yellow). (b)(c) Bar plots showing measured expectation values after preparing the 5-spin GHZ state(b) and 7-spin GHZ state (c). The colours indicate the number of qubits for which the measurement basis is not identity. Grey bars show ideal expectation values. (d) Plot of GHZ state fidelity against the number of constituent qubits. A value above 0.5 shows genuine N-qubit entanglement. The blue points are measured data, while the green points are theoretical predictions assuming a simple depolarising noise model whose parameters are extracted from single- and two-qubit experiments. [7]

5 DiVicenzo Criteria

The DiVincenzo criteria are a set of seven requirements essential for the realisation of a scalable quantum computer. Five of these criteria are fundamental for quantum computation, whilst the remaining two are specific to quantum communication. Below, we analyse how the NV centre in diamond addresses each of the five computational criteria.

1. **Scalability with well-characterised qubits**
   The NV centre satisfies this criterion by combining the electron spin with multiple nearby nuclear spins in the diamond lattice. As demonstrated in Section 3, a 10-qubit register has been successfully demonstrated, consisting of a single NV electron spin coupled to the intrinsic ^{14}N nuclear spin and eight surrounding ^{13}C nuclear spins. Each qubit is well-characterised through:

   - Distinct hyperfine coupling constants that allow individual addressing ($A_{\text{hf}}^{^{14}N} = 2.16$ MHz and $A_{\text{hf}}^{^{15}N} = 3.05$ MHz)
   - Spatial separation enabling selective control via frequency-selective RF pulses
   - Long coherence times suitable for quantum operations

   However, scalability beyond 10 qubits remains challenging due to increased crosstalk and decoherence, presenting an area for future development.

2. **Initialising qubits to a simple fiducial state**
   The NV centre achieves high-fidelity initialisation through optical pumping. As detailed in Section 2.2, the spin-selective inter-system crossing from the ^3E excited state preferentially decays to the $m_s = 0$ ground state sublevel. Through repeated cycles of green laser excitation at $\lambda = 532$ nm, the electron spin is polarised into the $|m_s = 0\rangle$ state with near-unity fidelity in less than 1 μs.

   Nuclear spins coupled to the electron are then initialised through electron-mediated polarisation via controlled gates, achieving high-fidelity initialisation of the entire multi-qubit register.

3. **Long relevant coherence times**
   The NV centre demonstrates excellent coherence properties:

   - **Electron spin T_1:** Longitudinal relaxation time of $T_1 = 5.4 \pm 0.5$ ms (Section 2.3.2), limited primarily by phonon interactions with the diamond lattice.
   - **Electron spin T_2^*:** Dephasing time of $T_2^* = 456 \pm 45$ ns (Section 2.3.4), limited by magnetic noise from the spin bath (primarily ^{13}C nuclear spins).
   - **Electron spin T_2^{echo}:** Coherence time with dynamical decoupling of $T_2^{\text{echo}} = 3.8 \pm 0.2$ μs (Section 2.3.5), achievable through Hahn echo sequences that suppress slow environmental fluctuations.
   - **Nuclear spin coherence:** Nuclear spins exhibit significantly longer coherence times, with measurements reaching tens of seconds when properly decoupled from the electron spin.

   The use of isotopic engineering (removing ^{13}C nuclei) can extend T_2 by up to two orders of magnitude, further improving coherence. Additionally, dynamical decoupling sequences and shaped pulses provide further mitigation of dephasing and crosstalk, enhancing the overall quantum coherence properties of the system.

4. **A "universal" set of quantum gates**
   A quantum gate set is considered universal if it can approximate any unitary operation on n qubits to arbitrary precision. For practical quantum computing, this requirement can be satisfied through the Solovay-Kitaev theorem, which states that any single-qubit unitary can be approximated using a finite set of discrete gates.

   The minimal requirements for universality are:

(a) **A complete set of single-qubit rotations:** Arbitrary rotations on the Bloch sphere, typically generated by rotations around two non-parallel axes (e.g., R_x and R_z rotations). These allow the construction of any single-qubit unitary.

(b) **At least one two-qubit entangling gate:** A gate that creates entanglement between two qubits, such as the CNOT gate, CZ gate, or iSWAP gate. This enables multi-qubit operations beyond the reach of single-qubit gates alone.

Alternatively, universality can be achieved using the **Clifford group** supplemented with a non-Clifford gate (such as the T gate or $\pi/8$ gate). The Clifford group consists of all single-qubit and two-qubit gates that map Pauli operators to Pauli operators. Whilst the group alone can only generate stabiliser codes and cannot perform universal computation, adding a single non-Clifford gate enables the approximation of any unitary operation through the Solovay-Kitaev decomposition. [12]

For the NV centre, the gate set comprising:

- Arbitrary electron and nuclear single-qubit rotations (via RF pulses)
- DDRF two-qubit gates realising conditional rotations between electron and nuclear spins

satisfies the requirements for universality. The high gate fidelities ($\mathcal{F}_{\text{gate}} = 0.991$) ensure that sequences of gates can approximate arbitrary unitaries within acceptable error tolerances for quantum error correction.

5. **A qubit-specific measurement capability**
The NV centre achieves qubit-specific measurement through optically detected magnetic resonance (ODMR), as detailed in Section 2.3.1. The spin-dependent PL provides a direct readout mechanism:

- **Electron spin readout:** The electron spin state is read by monitoring the PL intensity when the $m_s = 0$ state produces significantly higher PL intensity than $m_s = \pm 1$ states. A single microwave pulse initialises the system to a superposition, and optical readout measures the spin state with fidelity > 0.9.
- **Nuclear spin readout:** Nuclear spins are measured indirectly through their effect on the electron spin via conditional electron-nuclear gates. By adapting the measurement basis through nitrogen nucleus rotation (Section 3.3), arbitrary basis measurements can be performed, enabling full quantum state tomography.
- **Single-qubit resolution:** The hyperfine interaction provides qubit-specific coupling, allowing selective addressing and measurement of individual nuclear spins through frequency-selective RF pulses.

The spin-selective nature of the inter-system crossing and the high optical contrast between spin states enable measurement fidelities approaching unity, satisfying this criterion.

6 Conclusion

The nitrogen-vacancy centre in diamond represents one of the most mature and promising solid-state platforms for quantum information processing. This review has demonstrated that the NV centre successfully satisfies all five DiVicenzo criteria for quantum computation: it provides scalable, well-characterised qubits; enables high-fidelity initialisation to a fiducial state; exhibits long coherence times amenable to quantum error correction; supports a universal set of quantum gates; and permits qubit-specific measurements with near-unity fidelity. These capabilities collectively establish the NV centre as a viable candidate for practical quantum computing.

The key strengths of the platform are evident. Room-temperature operation eliminates the need for expensive cryogenic infrastructure, a major practical advantage over superconducting qubits. The spin-dependent photoluminescence enables direct, high-fidelity optical readout without complex measurement protocols. Electron coherence times ($T_2^{\text{echo}} \sim 3.8$ μs in bulk diamond) can be extended to tens of seconds

in purified, isotopically engineered samples [2], providing adequate timescales for meaningful quantum algorithms. The demonstrated 10-qubit register with verified 7-qubit entanglement confirms that multi-qubit scaling is achievable, and the DDRF gate protocol with fidelities exceeding 99% [7] provides a pathway to constructing larger quantum processors.

However, significant challenges remain before industrial-scale quantum computers based on NV centres become realisable. Scaling beyond 10 qubits faces mounting difficulties due to increased crosstalk between qubits and the exponential growth of circuit complexity. The limited connectivity of each NV centre (typically 8–10 controllable nuclear spins) restricts the qubit density per defect; realising large processors will require networks of multiple NV centres linked via alternative interfaces. Furthermore, whilst gate fidelities of 99.1% are impressive, fault-tolerant quantum computing is estimated to require fidelities exceeding 99.9% to suppress errors below the quantum error correction threshold. Achieving this requires further innovations in pulse shaping, composite gates, and environmental isolation.

Future research directions are clear. Isotopic engineering (removing ^{13}C nuclei and using ^{15}N) has proven effective for extending coherence; wider adoption of such techniques could improve T_2.

In conclusion, whilst the NV centre has proven its fundamental suitability for quantum computing, the path to a practical, scalable quantum computer remains open. The platform has progressed from proof-of-concept demonstrations to devices performing genuine multi-qubit quantum operations. Regardless, the NV centre's unique combination of room-temperature operation, optical control and readout, long coherence times, and demonstrated scalability ensures its continued importance as a test bed for quantum computing principles and a platform for near-term quantum sensing and simulation applications.

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